



SRI KRISHNA COLLEGE OF TECHNOLOGY

(An Autonomous Institution)

Affiliated to Anna University and Approved by AICTE
KOVAIPUDUR, COIMBATORE – 641 042.



LOAN ORIGATION AND CREDIT ASSESSMENT SYSTEM (LOCAS)

A PROJECT REPORT

23CS503 – APPLICATION DEVELOPMENT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

SRI KRISHNA COLLEGE OF TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with 'A' Grade
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SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed as a blueprint to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, were agreed upon by 193 nations to change the world for the better. The project is based on the following global goals:

QUESTIONS	ANSWER SAMPLES
Which SDGs does the project directly address?	SDG 8: Decent Work and Economic Growth & SDG 9: Industry, Innovation and Infrastructure – by accelerating credit access for MSMEs and individuals through automated risk scoring and paperless digital lending.
What strategies or actions are being implemented to achieve these goals?	End-to-end digital loan origination platform with automated FOIR/LTV calculators, real-time credit bureau pulls (CIBIL/Experian), Maker-Checker approval governance, and paperless e-sign workflows.
How is progress measured and reported in relation to the SDGs?	Tracking credit decision turnaround time (TAT reduction), paperless documentation compliance, credit default (NPA) rates, and tamper-proof SHA-256 audit trails.
How were these goals identified as relevant to the project's objectives?	Eliminates manual paper bottlenecks, human underwriting bias, credit miscalculations, and delayed loan disbursements to foster inclusive financial growth.
Are there any partnerships or collaborations in place to enhance this impact?	Collaborating with commercial retail banks, NBFCs, credit information bureaus (CIBIL, Experian, Equifax, CRIF), and financial regulatory bodies (RBI).

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**BONAFIDE CERTIFICATE**

Certified that this project report titled “**LOAN ORIGINATION AND CREDIT ASSESSMENT SYSTEM (LOCAS)**” is the Bonafide work of **ABISHEK B (727824TUIT004)** who carried out the project work under my supervision.

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INTERNAL EXAMINER

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ABSTRACT

Modern retail and enterprise banking operations demand strict regulatory compliance, accurate financial risk modeling, data security, and rapid credit decisioning across the loan lifecycle. Traditional credit underwriting processes frequently rely on manual paper applications, physical document verification, fragmented spreadsheets, and slow multi-tier approval channels. These outdated practices lead to administrative bottlenecks, human calculation errors, data inconsistencies, high turn-around time (TAT), and increased non-performing asset (NPA) risks. To eliminate these challenges, this project presents a full-stack, enterprise-grade **Loan Origination and Credit Assessment System (LOCAS)** developed using **Spring Boot 3, React 18, and MySQL 8**, engineered to automate, centralize, and streamline retail and business lending operations.

The platform delivers role-based dashboards tailored for key banking stakeholders, including **Applicants, Relationship Managers, Credit Officers, Field Verifiers, Credit Heads, and System Administrators**. Core capabilities include digital customer onboarding, automated KYC verification with hashed Aadhaar and masked PAN credentials, multi-bureau credit score pulls (CIBIL, Experian, Equifax, CRIF) requiring explicit applicant consent, automated FOIR (Fixed Obligation to Income Ratio) and LTV (Loan to Value Ratio) financial calculation engines, dynamic underwriting scorecards, and a mandatory **Maker-Checker governance model** enforcing Credit Head approval for high-value loans above INR 10 Lakhs.

By eliminating manual underwriting delays, enforcing strict credit policy rules, and incorporating SHA-256 tamper-evident audit logging, LOCAS enhances risk mitigation, regulatory compliance, operational efficiency, and customer satisfaction. The system provides a scalable digital foundation for modern banking institutions, drastically reducing loan processing times while maintaining audit-ready financial integrity.

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LIST OF ABBREVIATIONS

S. NO	ABBREVIATION	EXPANSION
1	LOCAS	Loan Origination and Credit Assessment System
2	FOIR	Fixed Obligation to Income Ratio
3	LTV	Loan to Value Ratio
4	CIBIL	Credit Information Bureau (India) Limited
5	KYC	Know Your Customer
6	DPD	Days Past Due
7	NPA	Non-Performing Asset
8	RBI	Reserve Bank of India
9	RBAC	Role-Based Access Control
10	JWT	JSON Web Token
11	REST	Representational State Transfer
12	API	Application Programming Interface
13	JPA	Java Persistence API
14	SQL	Structured Query Language
15	DFD	Data Flow Diagram

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid digital transformation of banking and financial services has drastically increased the demand for automated, secure, and transparent credit decisioning platforms. Traditional loan origination methods heavily depend on physical paperwork, manual document verification, isolated spreadsheets, and protracted multi-level approval hierarchies. These legacy approaches result in operational delays, high turn-around times (TAT), manual underwriting errors, and elevated non-performing asset (NPA) risks, burdening credit officers and creating frustration for retail and MSME loan applicants.

The **Loan Origination and Credit Assessment System (LOCAS)** is an enterprise platform designed to resolve these inefficiencies by delivering a centralized, secure, and regulatory-compliant digital lending ecosystem. LOCAS automates the end-to-end lending lifecycle—from initial customer registration, loan product selection, and automated financial ratio calculations (FOIR and LTV) to real-time credit bureau pulls (CIBIL, Experian, Equifax, CRIF), credit risk scorecards, field verification workflows, and Maker-Checker underwriting approvals.

Adhering strictly to financial regulations and banking security guidelines, LOCAS enforces stateless **JSON Web Token (JWT)** authentication, BCrypt password hashing, sensitive data encryption (AES-256 for Aadhaar hashing and PAN masking), and SHA-256 hash-chained audit logging. Furthermore, credit decisioning above configured financial thresholds (e.g., loans exceeding INR 10 Lakhs) strictly enforces a Maker-Checker policy requiring Credit Head approval. By eliminating manual bottlenecks and providing transparent, auditable credit decisioning, LOCAS establishes a robust, modern foundation for digital banking operations.

1.2 COMPONENTS OF SYSTEM

Role-Specific Dashboards: Tailored portals for Applicants, Credit Officers, Credit Heads, Field Verifiers, and Admins.

Credit Policy Engine: Manages product rules across HOME, PERSONAL, VEHICLE, EDUCATION, BUSINESS, GOLD, and LAP loans.

Financial Calculator Engine: Computes real-time FOIR, LTV, EMI, and max eligible loan amounts.

1.3 ADVANCED TECHNOLOGIES

Stateless JWT & Spring Security 6: Enforces granular Role-Based Access Control (RBAC) across endpoints.

Maker-Checker High-Value Threshold: Restricts single-officer approvals for loans above INR 10 Lakhs.

Explicit Customer Consent Bureau Fetch: Enforces data privacy compliance prior to pulling CIBIL/Experian credit scores.

SHA-256 Audit Logging: Maintains an immutable, insert-only audit log of all financial decisions and user actions.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM & DRAWBACKS

Traditional loan origination relies heavily on manual paper files, physical branch visits, manual FOIR calculations, and unencrypted spreadsheet records. Drawbacks include high turnaround time (TAT), elevated risk of human error, lack of verifiable audit trails, and unauthorized access to sensitive financial credentials.

2.2 PROBLEM DEFINITION & 2.3 PROPOSED SYSTEM

LOCAS solves these problems by delivering an automated, web-based digital platform built on Spring Boot 3, React 18, and MySQL 8. It features automated financial calculators, multi-bureau integration with explicit consent verification, role-based security, and dual-control Maker-Checker governance for loans above INR 10 Lakhs.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE & 3.2 SOFTWARE REQUIREMENTS

Hardware: Intel i5/i7 Processor, 8GB/16GB RAM, 512GB SSD.

Software Stack: Windows 11 / macOS, VS Code, ReactJS 18, Spring Boot 3, MySQL 8.



Fig. 3.1 - 3.4 Technology Stack Logos

CHAPTER 4

SYSTEM DESIGN

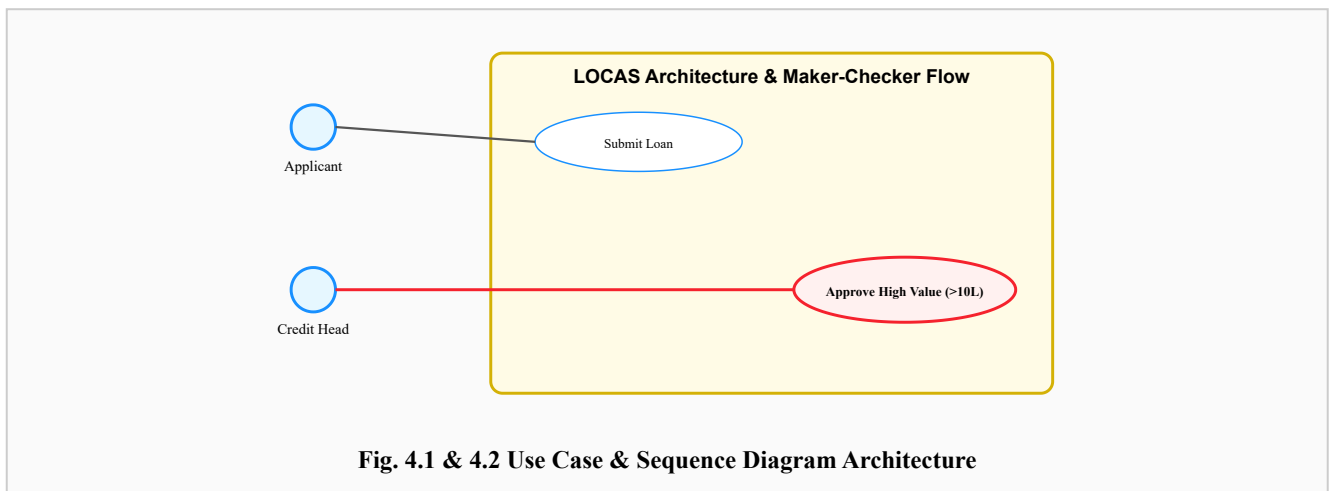
4.1 MODULE DESCRIPTION

LOCAS comprises 6 modules: User & RBAC, Credit Policy Engine, Applicant Onboarding, Underwriting & Scorecard, Credit Risk & Maker-Checker, and Bureau Integration & Audit Logging.

Table 4.1.5 Maker-Checker Threshold Policy Matrix

LOAN AMOUNT RANGE	REQUIRED AUTHORIZATION ROLE	ALLOWED DECISION ACTIONS
Up to INR 1,000,000 (≤ 10 Lakhs)	Credit Officer / Admin	APPROVE, DECLINE, REQUEST_CLARIFICATION
Above INR 1,000,000 (> 10 Lakhs)	Credit Officer (Maker) → Credit Head (Checker)	RECOMMEND (Officer) → APPROVE / DECLINE (Head)

4.2 USE CASE & 4.3 SEQUENCE DIAGRAMS



CHAPTER 5

TESTING

5.1 UNIT & INTEGRATION TESTING

Unit tests run via JUnit 5 & Mockito test password hashing, FOIR formulas, and threshold validation logic in isolation.

Integration tests evaluate REST API controllers and MySQL database multi-table commits.

5.3 SECURITY & AUTHENTICATION TESTING

Stateless JWT verification validates token expiration, bearer authorization injection, and role-based endpoint permissions.

5.4 TEST CASES

5.4.1 Test Case I: RBAC Access Control

Attempts by unauthorized roles to invoke admin or approval endpoints return HTTP 403 Forbidden.

5.4.2 Test Case II: Bureau Consent & Maker-Checker Threshold

Bureau pulls missing `consentRecordId` fail validation. Credit Officers attempting to approve loans > 10 Lakhs are blocked with: *"Loans above INR 10 Lakh require Credit Head approval. Please submit as RECOMMEND."*

MAKER-CHECKER POLICY ENFORCED

"Loans above INR 10 Lakh require Credit Head approval. Please submit as RECOMMEND."

Fig. 5.4 Test Case II Maker-Checker Enforcement Confirmation

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The **Loan Origination and Credit Assessment System (LOCAS)** provides a complete, secure, and regulatory-compliant digital lending platform built on Spring Boot 3, React 18, and MySQL 8. It successfully automates credit decisioning, enforces Maker-Checker threshold rules, protects sensitive customer data, and maintains SHA-256 tamper-evident audit logs.

6.2 FUTURE WORK

Future enhancements include AI/ML credit default predictions, Account Aggregator (AA) framework integration, and smart contract automated disbursements.

CHAPTER 7

APPENDIX I: SOURCE CODE

UnderwritingService.java (Maker-Checker & Threshold Enforcement)

```
@Service
@RequiredArgsConstructor
public class UnderwritingService {

    private static final BigDecimal HIGH_VALUE_THRESHOLD = BigDecimal.valueOf(1000000.00); // 10 Lakhs

    @Transactional
    public ApplicationResponse processUnderwritingDecision(Long applicationId, Long userId,
UnderwritingDecisionRequest request) {
        LoanApplication application = applicationRepository.findById(applicationId)
            .orElseThrow(() -> new ResourceNotFoundException("Application not found"));

        User user = userRepository.findById(userId)
            .orElseThrow(() -> new ResourceNotFoundException("User not found"));

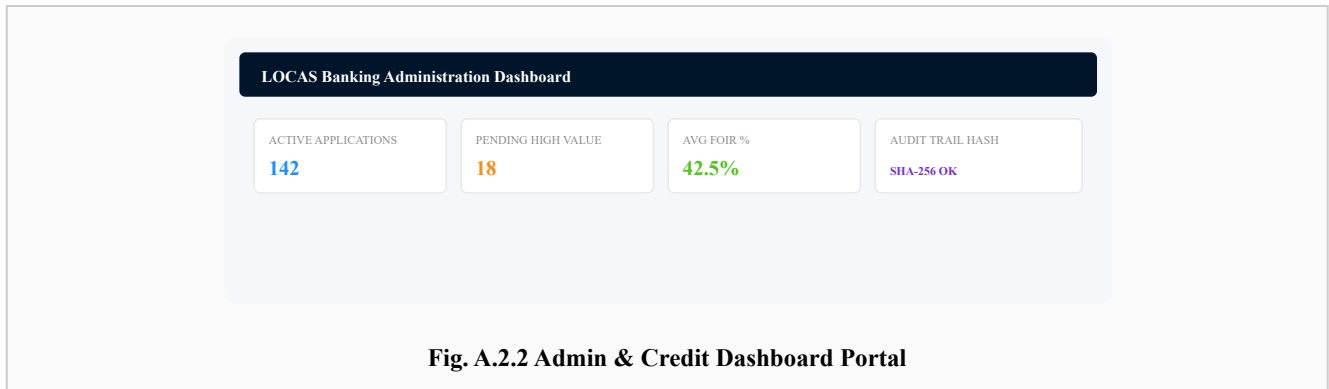
        // Maker-Checker Rule Enforcement: Loans above INR 10 Lakh require Credit Head approval
        boolean isHighValue = application.getRequestedAmount().compareTo(HIGH_VALUE_THRESHOLD) > 0;

        if (request.getDecisionType() == DecisionType.APPROVE) {
            if (isHighValue && user.getRole() == Role.CREDIT_OFFICER) {
                throw new ForbiddenException("Loans above INR 10 Lakh require Credit Head approval
(Maker-Checker policy). Please submit as RECOMMEND.");
            }
        }

        // Update Status & Save Decision...
        return mapToResponse(application);
    }
}
```

APPENDIX II & REFERENCES

A.2.1 User Login & Admin Dashboard Screenshots



REFERENCES

- [1] React Documentation: <https://react.dev/>
- [2] Spring Boot Documentation: <https://spring.io/projects/spring-boot>
- [3] Reserve Bank of India Digital Lending Mandates: <https://www.rbi.org.in/>